

Brightlight Tutorials - Data Analysis
SQL Fundamentals
Exercise 3 SQL Case statements

Table : products

product_id	product_name	price
1	Laptop	1200.00
2	Phone	800.00
3	keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

Q.1 Classify each product by price

- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

SELECT

~~product_id,~~
product_name,
price,

CASE

WHEN price > 1000 THEN 'Expensive'
WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
WHEN price < 100 THEN 'Budget'

FROM products;

Product Id	Product Name	Price	Price Category
1	Laptop	1200.00	Expensive
2	Phone	800.00	Mid-range
3	Keyboard	45.00	Budget
4	Monitor	300.00	Mid-range
5	Mouse	25.00	Budget

2 Table: Orders

Order Id	Customer Name	Amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Label each order:

- 'High Value' For orders ≥ 1000
- 'Medium Value' For 500 - 999.99
- 'Low Value' For orders < 500

SELECT

Customer_name,
amount,

CASE

WHEN Amount $\geq$ THEN 'High Value'

WHEN Amount BETWEEN 500 AND 999.99 THEN
'Medium Value'

WHEN amount < 500 THEN 'Low Value'

END AS orders_value

FROM order;

Customer name	Amount	Order_Value
Alice	150.00	Low Value
Bob	560.00	Medium
Charlie	999.99	Medium
Diana	45.50	Low
Ethan	1200.00	High

3 Table : employees

emp - Id	emp_name	department	Salary
1	john	IT	85000
2	sara	HR	60000
3	Mark	IT	75000
4	lucy	finance	45000
5	Tom	HR	55000

Question

Categorize employee position :

- IF in 'IT' and salary > 80000 → 'Senior IT'
- IF in 'HR' and salary > 55000 → 'Experienced HR'
- otherwise - 'staff'

SELECT

emp_name,
department,
Salary,

CASE

WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'

WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'

ELSE 'staff'

END AS position_category

FROM employees

emp_name	department	Salary	Position - Category
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4 Table : students

Student - Id	Student - name	Score
1	Anna	912
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Question :

Assign a letter grade:

- ≥ 90 : 'A'
- $80 - 89$: 'B'
- $70 - 79$: 'C'
- $60 - 69$: 'D'
- < 60 : 'F'

SELECT

Student - name
Score,

CASE

WHEN SCORE ≥ 90 THEN 'A'
 WHEN SCORE BETWEEN 80 AND 89 THEN 'B'
 WHEN SCORE BETWEEN 70 AND 79 THEN 'C'
 WHEN SCORE BETWEEN 60 AND 69 THEN 'D'
 ELSE 'F'

FROM students;

Student-Name	Score	Grade
ANITA	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

S. Table : deliveries

delivery - Id	delivery - time - minutes
1	45
2	80
3	30
4	65
5	100

Question :

Label delivery performance

- ≤ 30 mins : 'Fast'
- 31 - 60 mins : 'on Time'
- 60 mins : 'Late'

SELECT

delivery 'd',

delivery - time - minutes,

CASE

WHEN delivery - time - minutes ≤ 30 THEN 'Fast'

WHEN delivery - time - minutes BETWEEN 31 AND 60 THEN 'on Time'

~~WHEN~~ ELSE 'Late'

END AS performance

FROM Deliveries;

Results

delivery - Id	delivery - time - minutes	performance
1	45	ON Time
2	80	Late
3	30	fast
4	65	Late
5	100	Late

6 Table : tickets

ticket - Id	issue - type	priority
1	Login issue	1
2	Server down	3
3	Slow System	2
4	Email error	2
5	Password reset	1

Question :

Convert priority to label

- 3 → 'High'
- 2 → 'medium'
- 1 → 'Low'

SELECT

ticket - Id ,
issue type ,
priority ,

CASE

WHEN priority = 3 THEN 'High'
WHEN priority = 2 THEN 'medium'
WHEN priority = 1 THEN 'Low'

END AS priority - Label
FROM tickets ,

Results:

ticket - Id	issue-type	Priority	Priority-label
1	Login issue	1	Low
2	server down	3	High
3	Slow system	2	medium
4	Email error	2	medium
5	password reset	1	Low

7 Table: attendance

Student Id	days-present	total-days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Question:

Calculate attendance % and Classify:

- $\geq 90\%$ $\rightarrow$ 'excellent'
- $75 - 89\%$ $\rightarrow$ 'Good'
- $< 75\%$ $\rightarrow$ 'need improvement'

SELECT

Student - Id,

(day - present * 100.0 / total-days) AS attendance-
percentage,

CASE

WHEN (days-present * 100.0 / total-days) > 90 THEN 'excellent'
WHEN (days-present * 100.0 / total-days) BETWEEN 75 AND 89 THEN
'Good'

ELSE AS attendance-status

FROM Attendance;

Results:

Student Id	attendance - percentage	attendance - status
1	90.00	Excellent
2	60.00	Needs Improvement
3	96.00	Excellent
4	50.00	Needs Improvement
5	100.00	Excellent

8. Table: products - inventory

product - id	stock - qty
1	5
2	0
3	25
4	10
5	3

Question:

Label stock status:

- 0 → 'Out of stock'
- 1-5 → 'Low Stock'
- 5 → 'In Stock'

SELECT

product - id,

stock - qty,

CASE

WHEN stock - qty = 0 THEN 'Out of Stock'

WHEN stock - qty = 5 THEN 'In Stock'

WHEN stock - qty BETWEEN 1 AND 4 THEN 'Low Stock'

ELSE AS stock - status

FROM products - inventory

Result:

Product Id	Stock Qty	Stock Status
1	5	in stock
2	0	out of stock
3	25	in stock
4	10	in stock
5	3	low stock

Q. Table: Classes

Class Id	Subject	enrolled - Students
1	Math	30
2	English	25
3	Science	15
4	ART	5
5	History	20

Question:

Classify by size

- $\geq 25 \rightarrow$ 'Large'
- $10 - 24 \rightarrow$ 'medium'
- $< 10 \rightarrow$ 'Small'

```
SELECT subject,
        enrolled - students,
```

```
CASE
```

```
WHEN enrolled - students  $\geq 25$  THEN 'Large'
```

```
WHEN enrolled - students BETWEEN 10 AND 24 THEN 'Medium'
```

```
ELSE 'Small'
```

```
END AS class_size_category
```

```
FROM classes;
```

Results

Subject	enrolled_students	Class size - category
math	30	Large
English	25	Large
Science	15	medium
ART	5	Small
History	20	medium

10. Table: payments

payment_id	amount	payment_method
1	50.00	Card
2	200.00	Card
3	150.00	Card
4	75.00	paypal
5	300.00	Cash

Question:

Apply discount flag:

- IF payment_method = 'cash' and amount ≥ 200
- 'Eligible for discount'
- otherwise $\rightarrow$ 'not eligible'

```
SELECT payment_id,
       payment_method,
       amount,
```

CASE

```
WHEN payment_method = 'Cash' AND Amount >=
200 THEN 'Eligible for Discount'
ELSE Not Not Eligible
END AS discount_eligibility
FROM payments;
```

Result >

Payment ID	Payment method	amount	discount eligibility
1	card	50.00	Not Eligible
2	Cash	200.00	Eligible for discount
3	card	150.00	Not Eligible
4	paypal	75.00	Not Eligible
5	Cash	300.00	Eligible for discount